

TLA+ Study Group — Week 02

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November 30, 2025

Brief Description of the Topics

Lecture 4

- Axiomatizing the Die Hard 3 water jug puzzle
- TLA+ formulas must specify changes of "the entire world" i.e changes of all variables must be specified
- Rule of thumb: only use a primed variable on the LHS of equality $v' = \dots$ and set membership $v' \in$ formulas

Lecture 5

- Wedding as an analogy for database transactions.
- From this perspective, the goal of a wedding is making sure both parties believe they're in relationship (=committed) or they both aborted and it should be impossible for them to disagree on the outcome.
- A "really modern wedding" – with multiple participants i.e a database transaction consisting of several resource managers.
- All RMs must agree if they commit or abort.

What stood out

Lecture 4

- No type declarations. Variables types are just sets. Invariants seem like the worse (at least computationally) option.
- The axiomatize / simulate steps are reminiscent of logic programming

Lecture 5

- $\forall - \backslash A$
- $\exists - \backslash E$
- syntax for array expressions $\rightarrow$ e.g $\text{sqr} == i \in 1 \dots 42 \rightarrow i * i$ i.e $\text{sqr}[i] = i * i$ for $i \in 1 \dots 42$

- EXCEPT construct `rmState EXCEPT ![r] = "prepared"` equivalent to

```
rmState' = [ s \in RM |-> IF S = r THEN "prepared"
              ELSE rmState[s]]
```

- „Be suspicious of success” is a downer quip, although the ability to investigate the state space save it somewhat

Open Questions